

Project Design Phase-II

Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	
Project Name	AI FAQ Assistant API
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a visual representation of the information flows within a system. It shows how data enters and leaves AI FAQ Assistant API, what changes the information, and where data is stored. The Level-1 diagram on the next page uses the Gane-Sarson notation:

- **External entities (double rectangles):** people or systems outside the API (Admin / Editor, Customer client app, Support Agent, Email Service, LLM & Embedding API).
- **Processes (rounded boxes):** 1.0 Authenticate & Authorise, 2.0 Manage FAQs, 3.0 Answer Question, 4.0 Handle Feedback & Escalation.
- **Data stores (open-ended boxes):** D1 Users & API Keys, D2 FAQs + Embeddings, D3 Conversations, D5 Feedback & Tickets (PostgreSQL) and D4 Answer Cache (Redis).
- **Data flows:** arrows labelled with the data that moves, such as FAQ content, questions, answers with sources, and feedback.

How data moves: an editor logs in (1.0) and adds FAQs (2.0), whose text is sent to the embedding API and saved with its vector in D2. A customer app asks a question (3.0), which checks the cache (D4), finds matching FAQs in D2, asks the LLM to write an answer from them, and returns the answer with sources. Feedback and escalations (4.0) are stored in D5 and passed to a support agent.

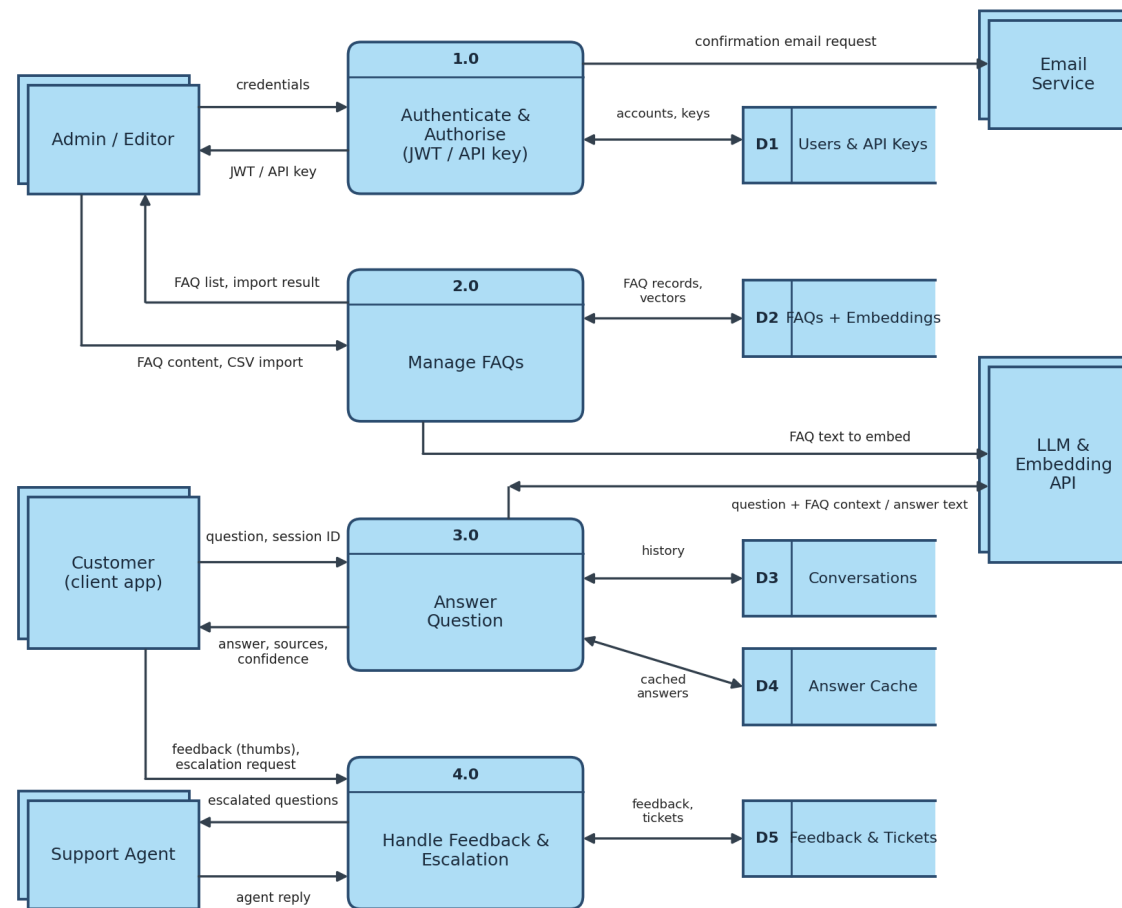


Figure 1: Level-1 Data Flow Diagram of AI FAQ Assistant API

User Stories

Use the below template to list all the user stories for the product. User story numbers, priorities and releases match the product backlog in the Project Planning document.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Developer	Project Setup	USN-1	As a developer, I can set up the NestJS project with environment config, PostgreSQL (with pgvector) and Docker so the team can build on a common base.	App starts with Docker Compose, connects to the database, pgvector is enabled and the health endpoint returns OK.	High	Sprint-1
Admin	Account & Access	USN-2	As an admin, I can register my organisation's account by entering email, password and confirm password.	Valid details create an account (201); duplicate email or mismatched passwords are rejected; password is stored hashed.	High	Sprint-1
Admin	Account & Access	USN-3	As an admin, I will receive a confirmation email once I have registered.	Email arrives with a working confirmation link.	Medium	Sprint-1
Admin	Account & Access	USN-4	As an admin, I can log in with email and password and receive JWT access and refresh tokens.	Correct credentials return both tokens; wrong credentials return 401.	High	Sprint-1
Admin	Account & Access	USN-5	As an admin, I can assign roles (Admin, Editor, Agent) and the API restricts endpoints by role.	Role change applies at next login; an Agent cannot edit FAQs (403).	High	Sprint-1
Editor	FAQ Management	USN-6	As an editor, I can create an FAQ with a question, answer and category (saved as a draft).	FAQ is saved as Draft with me as author; a missing question or answer returns 400.	High	Sprint-1
Editor	FAQ Management	USN-7	As an editor, I can view a list of FAQs and open a single FAQ.	List shows only my organisation's FAQs; an unknown ID returns 404.	High	Sprint-1
Editor	FAQ Management	USN-8	As an editor, I can update or delete an FAQ.	Changes are saved; a deleted FAQ is no longer returned or used in answers.	High	Sprint-1
Editor	FAQ Management	USN-9	As an editor, I can move an FAQ through Draft, Published and Archived states.	Status changes are saved; only Published FAQs are used to answer questions.	High	Sprint-2
Editor	FAQ Management	USN-10	As an editor, I can organise FAQs with categories and tags.	An FAQ has one category and several tags; FAQs can be filtered by them.	Medium	Sprint-2
Editor	FAQ Management	USN-11	As an editor, I can search, filter, sort and paginate FAQs.	Results match keywords and filters, follow the requested sort and include page details.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Developer	AI Search & Answering	USN-12	As a developer, I can connect an external LLM and embedding API through a provider-agnostic AI service module with prompt templates.	Provider can be changed by configuration only; calls use templates; timeouts and errors are handled.	High	Sprint-2
Developer	AI Search & Answering	USN-13	As a developer, I want an embedding created and stored in pgvector whenever an FAQ is created or updated.	New or edited FAQs have an up-to-date embedding; failures are retried and logged.	High	Sprint-2
Customer	AI Search & Answering	USN-14	As a customer, I can ask a question and get the most relevant FAQs ranked by meaning.	Top matches are returned with similarity scores even when wording differs; only Published FAQs are searched.	High	Sprint-2
Editor	FAQ Management	USN-15	As an editor, I can bulk import FAQs from a CSV or JSON file.	Valid rows are imported; invalid rows are reported with line numbers; embeddings are created.	Medium	Sprint-2
Customer	AI Search & Answering	USN-16	As a customer, I can ask a question and receive a short AI-written answer based only on matching FAQs, with source links.	Answer lists source FAQ IDs and contains no facts outside the matched FAQs.	High	Sprint-3
Customer	Feedback & Escalation	USN-17	As a customer, I get a confidence score, and a clear "I don't know" reply with an option to reach a human when no FAQ matches well.	Below the threshold the API returns a fallback reply and an escalation option, and does not invent an answer.	High	Sprint-3
Customer	AI Search & Answering	USN-18	As a customer, I can ask follow-up questions that use my earlier questions in the same session.	Follow-ups use session history; a new session starts fresh.	Medium	Sprint-3
Customer	Feedback & Escalation	USN-19	As a customer, I can give a thumbs up or down on an answer.	Feedback is saved with the answer ID; one vote per answer per session.	Medium	Sprint-3
Product Owner	Performance & Security	USN-20	As a product owner, I want repeated questions answered from a Redis cache to reduce cost and response time.	An identical question returns the cached answer without calling the LLM; the cache clears when FAQs change.	Medium	Sprint-3
Product Owner	Performance & Security	USN-21	As a product owner, I want rate limiting on question endpoints to prevent abuse and control cost.	Requests above the limit get 429 with retry information; the limit is configurable per API key.	High	Sprint-3
Developer	Documentation	USN-22	As a developer, I can read interactive Swagger (OpenAPI) documentation for every endpoint.	Swagger UI lists every endpoint with request and response schemas and accepts a JWT or API key.	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Support Agent	Feedback & Escalation	USN-23	As a support agent, I can view and answer escalated and unanswered questions.	The queue lists escalated questions with context; I can mark each as resolved.	Medium	Sprint-4
Admin	Analytics	USN-24	As an admin, I can see top questions, unanswered rate and satisfaction.	Report shows counts for the chosen dates and matches the stored data.	Medium	Sprint-4
Third-party Developer	Account & Access	USN-25	As a third-party developer, I can use an API key to access the API from my own application.	A valid key gives access to its own organisation's data only; an invalid or revoked key returns 401.	High	Sprint-4
Developer	Quality & Testing	USN-26	As a developer, I want unit tests for the auth, FAQ and AI service modules.	Unit tests for the three modules run and pass in the CI pipeline.	High	Sprint-4
Developer	Quality & Testing	USN-27	As a developer, I want end-to-end tests covering the main API flows.	End-to-end tests for register, FAQ, ask and escalation flows pass on a test database.	High	Sprint-4
Developer	Quality & Testing	USN-28	As a developer, I want global validation, exception filters and logging for consistent error responses.	All errors share one JSON format; requests and errors are logged.	Medium	Sprint-4
Developer	Deployment	USN-29	As a developer, I can build and test the project automatically through a CI/CD pipeline.	Every push runs build and tests; failing tests block the merge.	High	Sprint-4
Product Owner	Deployment	USN-30	As a product owner, I want the API deployed to a cloud environment with production configuration.	API is reachable over HTTPS and secrets are kept outside the code.	High	Sprint-4